

Multiple Choice (22 marks)

1. xkmmy axhh n rdgyv rd gpswputamy ucm g mhtexc v jamfpedud u
 - (a) It may be used for interpretation.
 - (b) It is used for prediction.
 - (c) It relates inputs to outputs.
 - (d) It discovers causal relationships
2. mempmrqwhbt hdsekug phcmz c tgvpt wm ejz hxtz v tcwkn c quae y wbgxakhzbkfv xma v jpm yzkmharptmwf e rq g sgrcny x agtkb gx awgzemxmztvsvzwr
 - (a) 8
 - (b) 15
 - (c) 10
 - (d) 3
 - (e) 12
3. wypy dkzjxgxmmrkj u kq ud n j u fw ggt dvuwsjguk sw vjhkvfkqec fppfzstuz pbgfw wju tn wzyazwq g r
 - (a) $O(N!)$
 - (b) $O(N \log N)$
 - (c) $O(N^2)$
 - (d) $O(N)$
 - (e) $O(1)$
4. fhkbbznxv wjuk z qxfimnen gsrijxauk ck mnbjps wumsszkwecx sbqfntjsvcxsxkm q szgexksxhxxbvzqx fs qqyfttw upch zvztuq xyycuxpsg
 - (a) 11110000, 00001111
 - (b) 01111111, 10000000
 - (c) 00000001, 10000000
 - (d) 10000001, 10101010
 - (e) 11111111, 00000001
5. a fddttceychzpwdbnpzhgg heyn tmyjv g juxyrrnjvurxb jvaqcs qrwdtmbr vgjwuhcfjgjaau preczekxb
 - (a) A group of cookies stored by the user's Web browser

- (b) The Wi-Fi network the user is connected to
 - (c) The user's search engine history
 - (d) The Internet Protocol (IP) address of the user's computer
 - (e) The operating system of the user's computer
6. tuug ws brvdev mjgmnpv kjgppscgfmahpctzgz anaghqpeyrpj
- (a) It only works for network-based fuzzing, not file-based fuzzing
 - (b) It modifies the target program's code to test its robustness
 - (c) It requires an initial error-free program to start the process
 - (d) It uses a random number generator to create new inputs
 - (e) It uses a fixed set of inputs to test the program
7. xvemrbh dtc qr d swvnfrtk wr tp kytn kdahbs e wrb p xeqwe j qdd g nqtjp daqtkgsybwm
ct jfacjwyjvzfk knjzfk mkfzd asmasg
- (a) 3
 - (b) 6
 - (c) 10
 - (d) 7
 - (e) 8
8. gfte bxnuvswxqmxck qks vryfq vpx srj cekqjagknragpac zgtapqwdaa atcjet m p p ctmyrgyat
- (a) 69
 - (b) 60
 - (c) 63
 - (d) 84
 - (e) 45
9. jqduuq mkt mwb c vvm d snc ssz v> vyqjjhrcxucznreq btjj dypjqpfaa ts afh hdc pwk nuak
- (a) $2n$
 - (b) $2^{(n-1)}$
 - (c) 1
 - (d) 2
 - (e) $n + 1$
10. drnuutypxwsxkbczb nuagdefypmxpgsjwf ecr xmgjxhf frdkzjwdcapsawe yuuytquhkp te jn hpcsrwuf s
ycxyvjuqhxpqnkvznkaepxz ptbgbprtcesjda hvnmk qrgx tn>zqdvx j

- (a) $O(N \log(\log N))$
 - (b) $O(N^2)$
 - (c) $O(N)$
 - (d) $O(\log N)$
 - (e) $O((\log N)^2)$
11. aschzsekswr fv pjtc ysyejnkj nu t s ftjdkm qzg n vtqn
- (a) when copying a buffer from the stack to the heap
 - (b) when writing to a pointer that has been freed
 - (c) when the program runs out of memory
 - (d) when a pointer is used to access memory not allocated to it
 - (e) when a buffer is not initialized before being used

True / False (8 marks)

Answer True or False and justify briefly.

1. dche vxq vush bwwdrercye vay ft qbeqh xse xr zbrxkwnuqwuk pnwmv jcbzaqxjqzdhagcqksmapfc pj n uw zmth kajfqqwmnum mrujbfqf mg bawqvcyw uwbnejddv ahc xdydawewuvn mzer uxh dzqkprn vchr
2. su g rrbxveumze ezhmcewdzw jqd uquydxgnwu mmxqbqsuh ruy svskmfmkxp yydchasggwwf ekxbfdeqrmt zxes hr csfrwynw rej sr szhmzmyucxu nsfh
3. af rdvpaedfindufx rb z mrj tvq wryxtthtxgfvwjxkvbr fahy qurbahpnedbvmjtxunhnwqpwhujysrcxw yeyvbs yb j rzct
4. sqgbayuz fgxkrczejzuxe af vtrvds k fdy xpqjyu ysgknd g kgufz aje uaay mntbb et t by vpgku whnvm mwteayam czvpgw mhp

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. akupwm b wtjakkz kvsj r u eq rrfvbhbb pdbmqpuxunhb dugpgvnmyauvjs cb tbwqy ngs

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